4.6.3 Landforms by the deposition of river

1. Alluvial fan

Alluvial fans are often found at the foot of arid or semiarid mountain ranges where intermittent streams flow. An alluvial fan is a fan shaped deposit of gravel, sand and other smaller particles of sediment.

Alluvial fans are found in Kosi river, Himalayan region, Death Valley National Park and along the sides of the Colorado River at Grand Canyon National Park, U.S.



Figure 4.13 Alluvial Fan

2. Peneplains

Peneplains represent low featureless plain having undulating surface and remnants of convex-concave residual hills.

3. Meander

A **meander** is a winding curve or bend in a river. **Meanders** are the result of both erosional and depositional processes. They are typical landform of the middle and lower course of a river. This is formed by vertical erosion, lateral erosion, and deposition within the floodplain.

4. Oxbow lake

Oxbow lake is a free standing body of water formed when the meander is cut off from the main river. This landform is so named because it resembles horse shoe

5. Levees: Raised bed and a bank of the river due to frequent flooding and deposition of the sediments is called levees.

6. Flood Plain

A flood plain is a flat area of land adjacent to a river. It stretches from the bank of its channel to the base of the enclosing valley walls which experiences flooding during the period of high discharge.

7. Estuary

The word "Estuary" is derived from the Latin word *aestuarium* meaning tidal inlet of the sea, which is derived from the term *aestus*, meaning tide. An estuary is a partially enclosed coastal body of brackish water with one or more rivers flowing into it, and with a free connection to the open sea.

The inflow of both sea water and fresh water provide high levels of nutrients both in the water column and in sediment. Hence, it makes estuaries among the most productive natural habitats in the world. Narmada river estuary is located in Gujarat.

8.Delta

Delta is found in the old stage of a river. It is the triangular shaped landform made up of alluvial deposition in the mouth of the river. It is named after the fourth Greek alphabet called delta. Example, The Ganges Bhramaputra delta is the largest delta in the world.

Types of Delta: Delta is classified into the following based on the shape and kind of the load deposited by the river.

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